Computer Science & IT

Theory of Computation



CFL

Lecture No. 6



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Recap of Previous Lecture



- Topic — CYK
- Topic — CNF, GNF
- Topic

Topics to be Covered



- Topic
- Topic
- Topic

GNF ✓
↓
CFG ✓
↓
String matching ✓

GNF

terminal

$A \rightarrow a\alpha$

$a \in \Sigma \quad \alpha \in V^*$

$A \rightarrow aBC/b$

$B \rightarrow bAC/a$

$C \rightarrow bCC/b$

GNF → PDA

$A \rightarrow \epsilon$

$A \rightarrow B$

CNF-X

GNF-X

|w| |w| steps

Conversion of GNF to PDA :-

1) Push the start symbol 'S' on to the stack

$$\delta(q_0, \epsilon, Z_0) \rightarrow (q_1, SZ_0)$$

2) Push RHS of S as follows

$$\delta(q_1, a, S) = \delta(q_1, \alpha)$$

if $S \rightarrow a\alpha$ is in 'G'

$$\delta(q_1, b, S) = \delta(q_1, \beta)$$

if $S \rightarrow b\beta$ is in G

3) add final state with $\delta(q_1, \epsilon, Z_0) = (q_f, Z_0)$

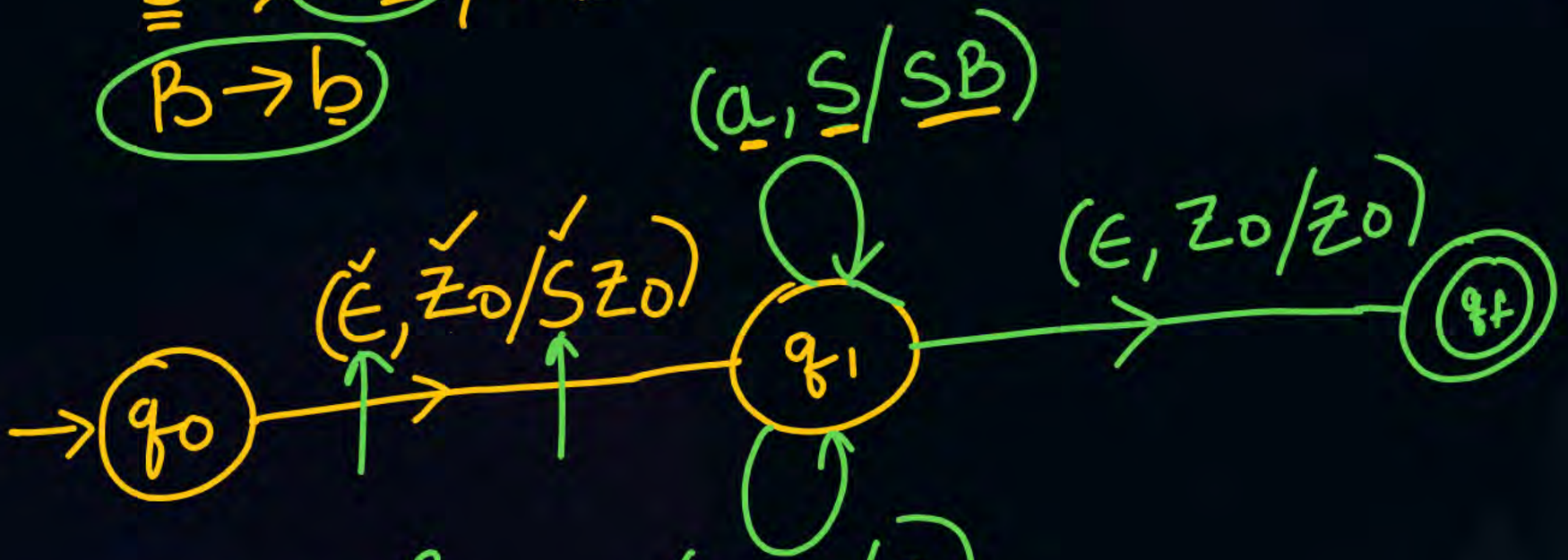
$S \to aSb/ab$

$S \to aSB/aB$

$B \to b$

$S \to aSB$





$S \to aB$

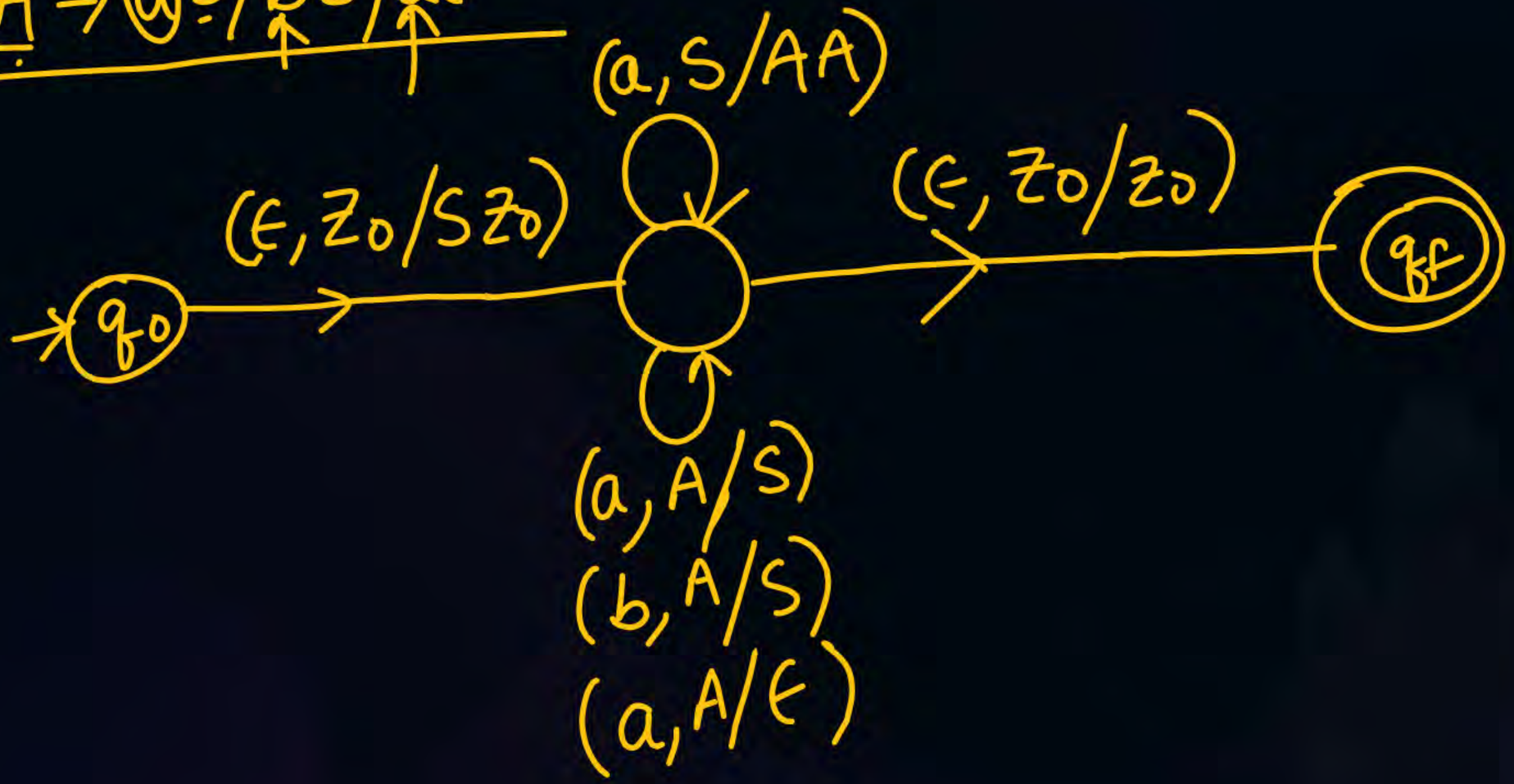
$B \to b\epsilon$

$(a, S/B)$

$(b, B/\epsilon)$

$S \rightarrow aAA$

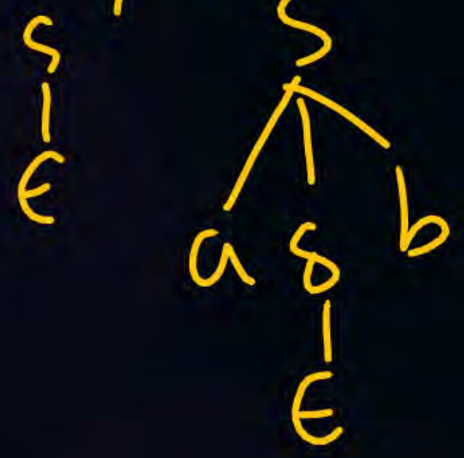
$A \rightarrow aS / bS / a\epsilon$



Context free grammars :-

$$L = \{a^n b^n / n \geq 0\}$$

$$S \rightarrow aSb / \epsilon$$





$L = \{a^m b^m c^n d^n / n, m \geq 0\}$

$S \rightarrow AB$

$A \rightarrow aAb/\epsilon$ $\quad A \overset{*}{\Rightarrow} a^n b^n$

$B \rightarrow cBd/\epsilon$ $\quad B \overset{*}{\Rightarrow} c^m d^m$

$a^n b^n c^n d^n \subseteq a^m b^m c^n d^n$

CSL $\qquad$ DCFL

$a^n b^n / n \geq 1$

$S \rightarrow aSb / ab$

$a^n b^n / n \geq 2$

$S \rightarrow aSb / aabb$

$a^n b^{2n} / n \geq 0 \rightarrow$ DCFL

$S \rightarrow aSbb / \epsilon$

What is the language

$S \rightarrow asb / abb$

$a^n S b^n$

$\downarrow$

abb

$a^{n+1} b^{n+2}$

$S \overset{*}{\Rightarrow} a^n S b^n$

$\downarrow$

$a^n abbb^n \Rightarrow a^{n+1} b^{n+2} / n \geq 0$

$a^m b^{m+1} / m \geq 1$ $\rightarrow$ DCFL

$S \rightarrow aSbb \mid aab$

$S \rightarrow aSbb$

$S \overset{*}{\Rightarrow} a^n S b^{2n}$

$\Big|$

aab

$a^n aabb^{2n}$

$= a^{n+2} b^{2n+1}$ / $n \geq 0$ → PCFL ✓

$a^{2n}b^n / n \geq 0$

$S \rightarrow aaSb / \epsilon$

$a^n b^{10n}$ → DCFL

$a^{10n} b^n$ → DCFL

$a^{100n} b^n$ → DCFL

$a^n b^{n+3} / n \geq 0$

$a^n bbb\, b^n$

$S \rightarrow aSb / bbb$

$a^n S\, b^n$

$|$

bbb

$a^n b^{2n+3} / n \geq 0$

$a^n bbb b^{2n}$

$S \rightarrow aSbb / bbb$

$a^n b^{3n+4}, n \geq 0$

$a^n bbbb b^{3n}$

$S \rightarrow a S bbb / bbbb$

$a^{3n+2}\, b^n / n \geq 1 \Rightarrow a^5 b$

$a^{3n}\, aa\, b^n$

$S \rightarrow aaaSb / aa / a^5 b$

$a^{2n}\, b^{3n} / n \geq 0$ → DCFL

$S \rightarrow aaSbbb / \epsilon$

2 a's → push 1a

3 b's → pop 1a

$a^{100n}\, b^{100n}$

$a^{100n}\, b^{2n}$

} DCFL

$a^{2n+3}\, b^{3n+2} / n \geq 0 \rightarrow$ DCFL

$\underline{\underline{a}}^{2n}\, aaa\, bb\, \underline{\underline{b}}^{3n}$

$S \rightarrow aaSbbb / aaabb$

$a^m b^n / m \geq n / n, m \geq 0$ DCFL

$a^* a^n b^n$

$S \to AB$

$B \to aBb / \epsilon$ $a^n b^n$

$A \to aA / \epsilon$ a^*

✓

$S \to Aasb /$

$A \to aA$

$a^m b^n / m \geq n$, $m, n \geq 0$ – DCFL

$a^+ a^n b^n$

$S \rightarrow BA$

$A \rightarrow aAb/\epsilon$ $a^n b^n$

$B \rightarrow aB/a$ a^+

$a^m b^n / m \leq n$ — DCFL

$a^n b^n b^*$

$S \rightarrow AB$

$A \rightarrow aAb / \epsilon$

$B \rightarrow bB / \epsilon$

$a^m b^n / m < n \rightarrow$ CFL

$a^n b^n b^+$

$S \rightarrow AB$

$A \rightarrow aAb / \epsilon$

$B \rightarrow bB / b$

$a^m b^n / m \neq n \rightarrow$ DCFL

$a^m b^n / m > n$

$a^+ a^n b^n$

$a^m b^n / m < n$

$a^n b^n b^+$

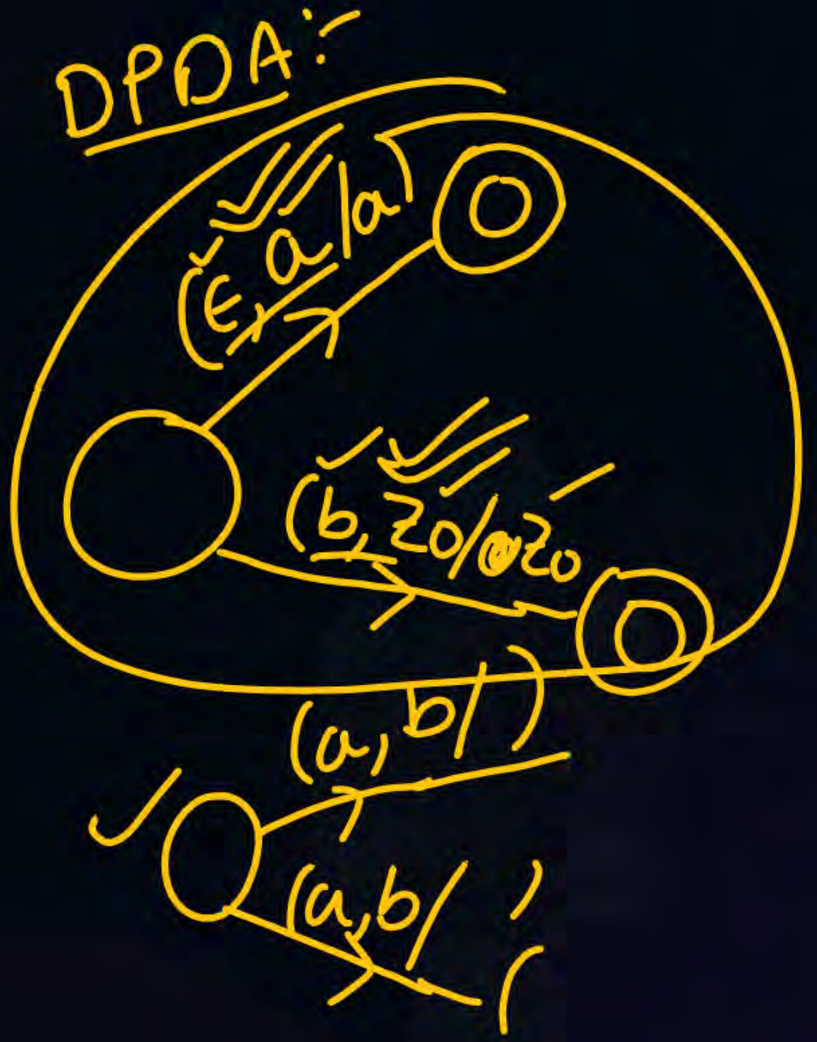
S → BA / AC

A → aAb / ϵ

B → aB / a

C → bC / b

DPDA :-



$S \to AS_1 / S_1B$

$S_1 \to aS_1b / \epsilon$ $\quad a^nb^n$

$A \to aA / \epsilon$ $\quad a^*$

$B \to bB / b$ $\quad b^+$

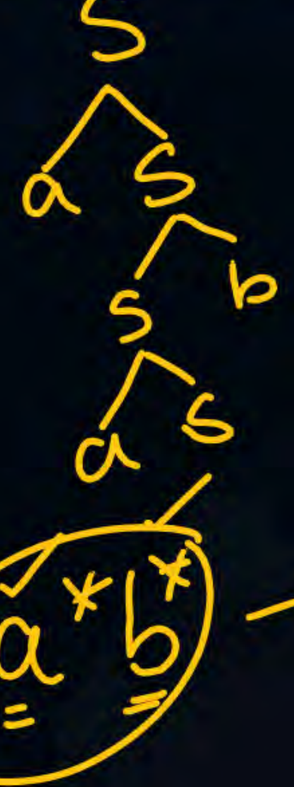
$a^*a^nb^n \cdot a^nb^nb^+$

$a^mb^n + a^mb^n$

$m \geq n$ $\quad m < n$

$m > n \quad m = n \quad m < n$

$S \to aS / Sb / \epsilon$



$a^*b^* \to$ Reg

$a^m b^n / m \neq 2n$ → DCFL

$a^* a^{2n} b^n$

$S \to BA$

$A \to aaAb/\epsilon$

$B \to aB/\epsilon$

$a^{2n} a^* b^n$

$S \to aaSb/B$ ✓

$B \to aB/\epsilon$

$a^{2n} S b$

↓

B

|

a^*

$a^m b^n / m \geq 2n+3$ → DCFL

$a \; a^{2n+3} \; b^n$

$2n+3**$

$S \to BA$

$A \to aaAb / aaa$

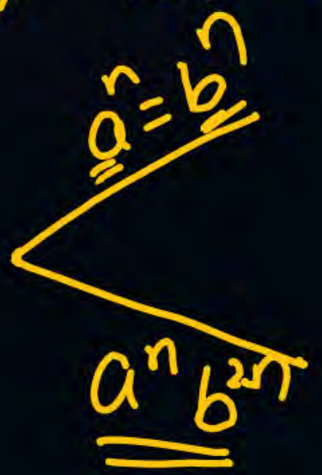
$B \to aB / \epsilon$

$a^m b^n / m \leq 2n+3$ → DCFL

CFG → difficult

$a^n b^n \cup a^n b^{2n}$ → DCFLX CFL✓

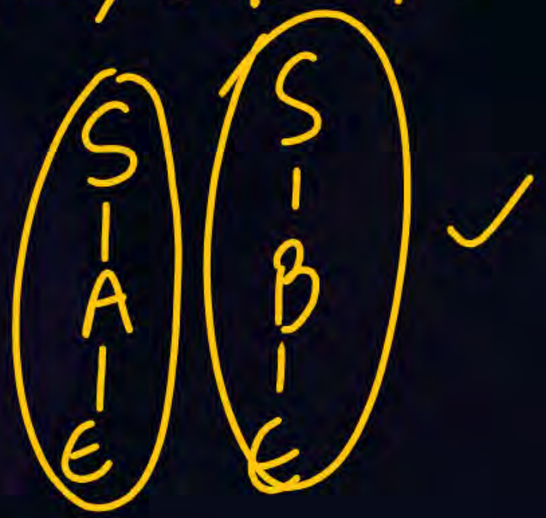
DPDAX NPDA✓



$S \to A/B$

$A \to aAb/\epsilon$

$B \to aBbb/\epsilon$



✓

* $a^m b^n / m \le n \le 2m \rightarrow$ NDPA $\rightarrow$ CFL

XDPDA $\rightarrow$ XDCFL

$2 \le n \le 4$

$S \rightarrow aSb / aSbb / \epsilon$ ✓

$\Sigma = \{a, b, c\}$

$a^m b^m c^n / m, n \geq 0$ → DCFL

$a^n b^n c^*$

$S \rightarrow AB$

$A \rightarrow aAb/\epsilon$

$B \rightarrow cB/\epsilon$

$a^* b^n c^n$

$a^m b^n c^n / m, n \geq 0 \longrightarrow$ DCFL

$S \to BA$

$A \to bAc / \epsilon$

$B \to aB / \epsilon$

$a^*(bc)^*$

$a\,bc\,bc$

$S \to aSbA / \epsilon$

$A \to cA / \epsilon$



$a^m b^n c^p$ / $m = n+p$ $\rightarrow$ DCFL $\quad m,n,p \geq 0$

$a^{n+p} b^n c^p$

$a^p (a^n b^n) c^p$

$S \rightarrow aSc / B$

$B \rightarrow aBb / \epsilon$





$a^m b^n c^p / p = m+n \longrightarrow$ DCFL

$a^m b^n c^{m+n}$

$a^m b^n c^n c^m$

$S \rightarrow aSc / A$

$A \rightarrow bAc / \epsilon$

$a^m b^n c^p / n = m+p \longrightarrow$ DCFL

$a^m\ b^{m+p}\ c^p$

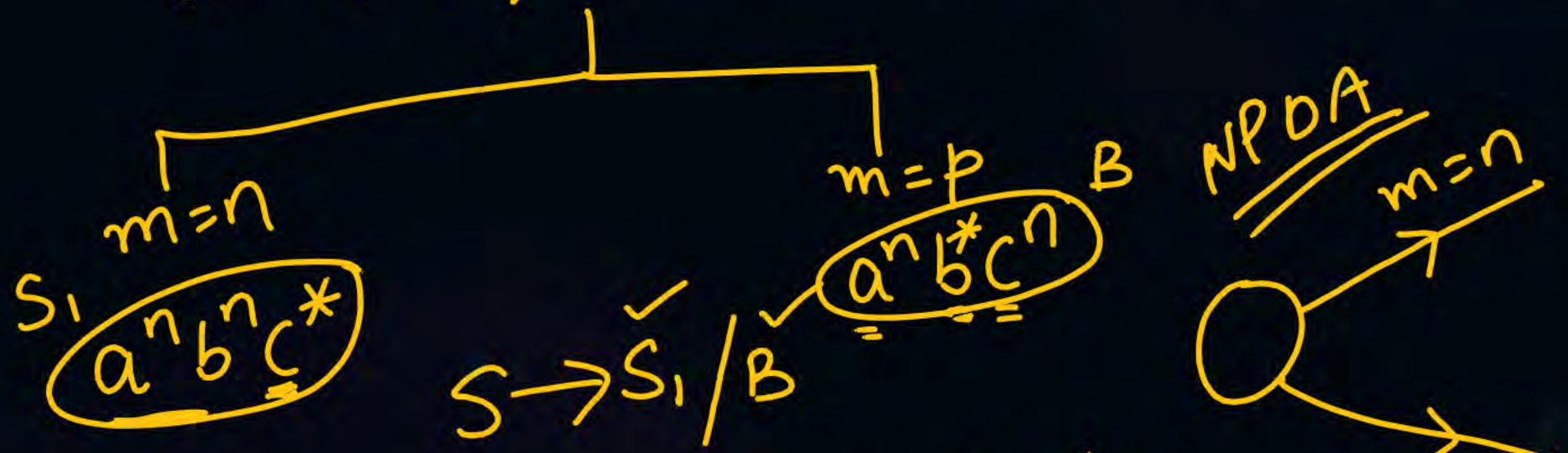
$a^m b^m b^p c^p$

$S \to AB$

$A \to aAb/\epsilon$

$B \to bBc/\epsilon$

$\{a^m b^n c^p / m=n \cap m=p\}$ → CFL XDCFL



S1 → AC
A → aAb/ϵ
C → cC/ϵ

B → aBC/D
D → bD/ϵ



$a^m b^n c^p d^q / m=n$ and $p=q$ → DCFL

$a^n b^n c^m d^m$

$S \rightarrow AB$

$A \rightarrow aAb/\epsilon$

$B \rightarrow cBd/\epsilon$

$a^m b^n c^p d^q$ / $m=p$ and $n=q$ → CSL ✓

✗ CFL

✗ Reg

✗ DCFL

$a^m b^n c^p d^q$ / $m+n=p+q$ → DCFL

Push Pop

CFG → difficult

$a^m b^n c^p d^q \mid m+q = n+p$ → DCFL

CFG → difficult

$a^m b^n c^p d^q$ / $m+p = n+q$ → DCFL ✓

CFG → difficult

$a^m b^n c^p d^q$ / $m = p$ and $n = q$ → CSL ✓ CFL ✗

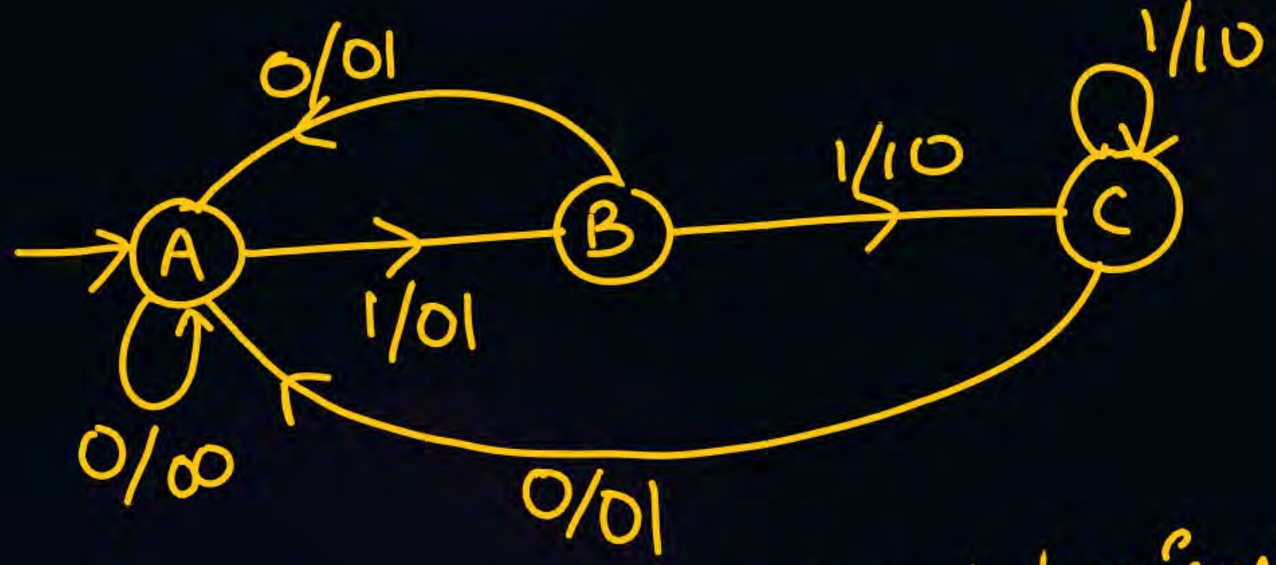


alternate addition in DCFL

alternate comp in CSL

PYQ

11:40



Sum of present and previous bits

The smallest FA which accepts the language

Consider the following NFA M, accepting language L
what is the language accepted by Complement of 'M'



(a) $L_1 = \{0,1\}^* - L$

b) $L_1 = \{0,1\}^*$

c) $L_1 \subseteq L$

d) $L_1 = L$



$L_1 = \{0,1\}^*$

$\epsilon, 0, 1, 00, 01, 10, 11$

Which of the following regular expressions is not equivalent to the reg exp $(a+b+c)^*$

a) $(a^*+b^*+c^*)^*$

b) $(a^*b^*c^*)^*$

c) $((a+b)^*+c^*)^*$

d) $(a^*(b^*+c^*))^*$

aa

$(a+b)^* = (a^*+b^*)^* = (a^*+b)^* = (a+b^*)^* = (a^*b^*)^*$

Let $M = (Q, \Sigma, \delta, q_0, F)$ be a finite automata

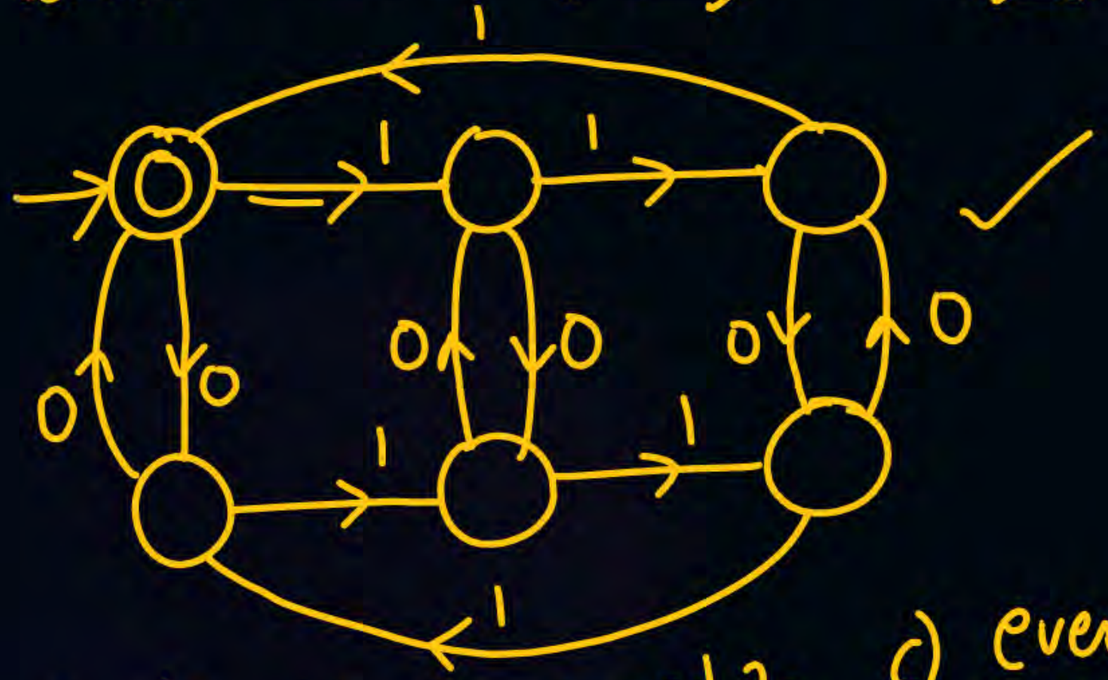
where $Q = \{A, B\}$ $\Sigma = \{a, b\}$ $q_0 = A$ $F = \{B\}$

$\delta(A, a) = A$, $\delta(A, b) = B$, $\delta(B, a) = B$ and $\delta(B, b) = A$

What is RLG?

$A \rightarrow aA$ $A \rightarrow bB$ $B \rightarrow aB$ $B \rightarrow bA$ $B \rightarrow \in$

The following FSM accepts all those binary strings in which number of 1's and 0's are respectively



a) divisible by 3 and 2
b) odd and even
c) even and odd
d) divisible by (2) and (3)

Which of the following statements is true about the regular expression 01^*0?

a) It represents a finite set of finite strings

b) " infinite set " "

c) " finite set of infinite strings

d) " infinite set of infinite strings

any string is always finite $1, 2, 10^{10}, 10^{10^{10}}$

$01^*0 \rightarrow$ infinite language

The language $\{0^n 1^n 2^n / 1 \le n \le 10^6\}$ is

finite

a) Regular ✓

b) ~~CFL~~ but not Reg ✗

c) Context free but its complement is not context free ✗

d) Not CFL ✗



→ Consider the following NFA



If Y is final state, then language is L_1

If Z is final state, then language is L_2

$Y = X0 + Y0 + Z1$

$Z = X0 + Y0 + Z1$

a) $L_1 = L_2$
b) $L_1 \subset L_2$
c) $L_2 \subset L_1$
d) None

THANK - YOU